

Books Data Extraction & Web Scraping

Data Analyst Portfolio Project Report

Author	Bramwel Mwasenga
Domain / Project Type	Web Scraping & Data Cleaning
Tools	Python, BeautifulSoup, Pandas, Excel, VS Code, Git/GitHub
Portfolio	https://bramwelsamwel16.github.io/portifolio/
GitHub	https://github.com/bramwelsamwel16

1. Project Overview

This project demonstrates a complete **web scraping** workflow: extracting data for 1,000 books from the practice website **books.toscrape.com** using Python, saving it to Excel, then cleaning and analyzing it to surface useful insights about book prices and ratings. The project falls under the **Data Extraction / Data Cleaning** category, which is a critical foundation before any deeper data analysis.

2. Business Problem

Companies and analysts often need data that only exists on a website rather than in a ready-made file (CSV/Excel). Key question: *How can we automatically extract product (book) data directly from a website, then transform it into clean, analysis-ready data that supports decisions — such as understanding price distribution or product quality (rating)?*

3. Dataset Information

Source	https://books.toscrape.com/catalogue/
Number of Records (Rows)	1,000 books
Number of Columns	4 (Title, Price, Rating, Availability)
Pages Scraped	50 pages (20 books per page)
File Format	.xlsx (Excel)

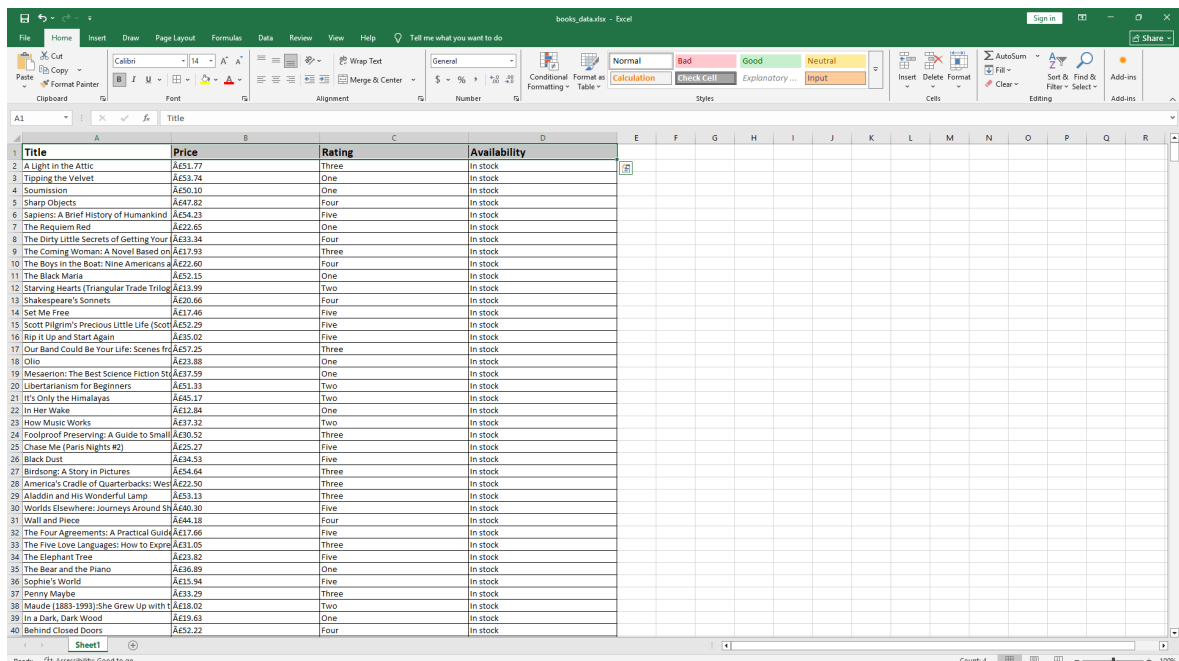
4. Tools & Technologies

- **Python 3** — the main language used to write the scraping script
- **requests** — downloads the HTML of each website page
- **BeautifulSoup4 (bs4)** — parses the HTML and extracts each book's details
- **Pandas** — stores the data in a DataFrame and writes it to Excel (.xlsx)
- **Visual Studio Code (VS Code)** — environment used to write and run the code
- **Microsoft Excel / openpyxl** — cleaning the data, building tables, formulas and summaries
- **Git & GitHub** — version control and publishing the project publicly
- **GitHub Pages** — hosting the portfolio online

Image 1: VS Code showing the scraper . py code (top) and terminal output (bottom) confirming that 1,000 books were extracted from 50 website pages, and that the Excel file was created successfully.

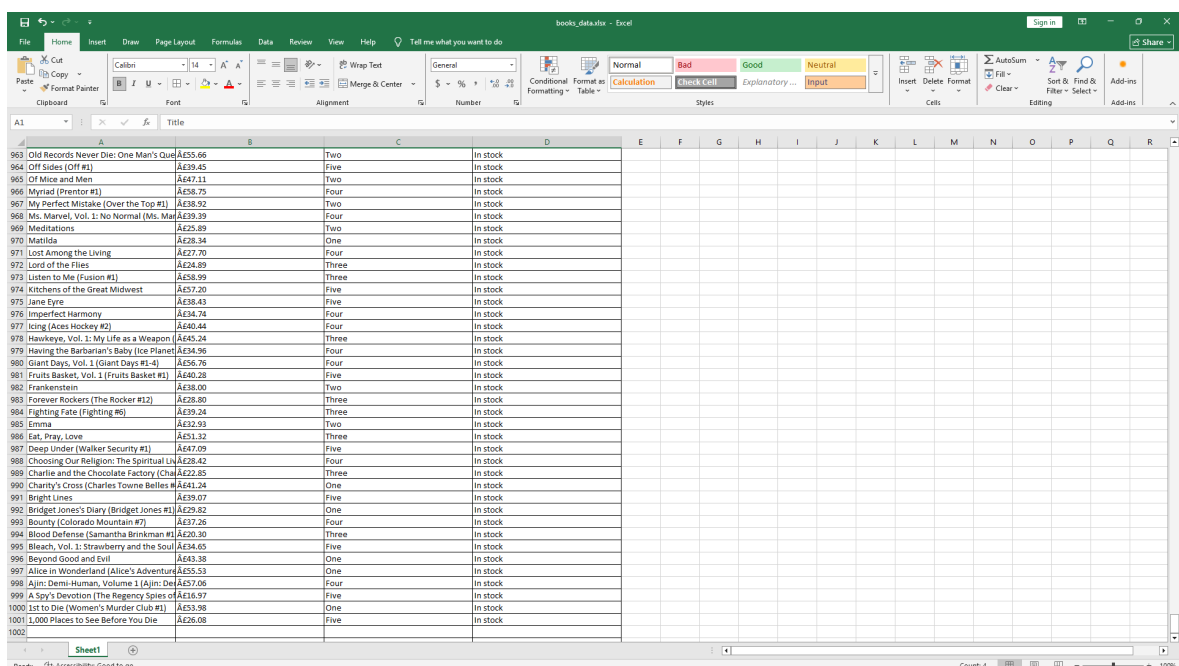
6. Data Cleaning & Preparation

Before cleaning, the raw data produced by the scraper had the following structure (4 columns: Title, Price, Rating, Availability) as shown in Images 2 and 3 (the data opened in Microsoft Excel):



	A	B	C	D
	Title	Price	Rating	Availability
1	A Light in the Attic	£51.77	Three	In stock
2	Tipping the Velvet	£53.74	One	In stock
3	Soumission	£50.10	One	In stock
4	Sharp Objects	£47.82	Four	In stock
5	Sapiens: A Brief History of Humankind	£54.23	Five	In stock
6	The Requiem Red	£22.65	One	In stock
7	The Dirty Little Secrets of Getting Your	£33.34	Four	In stock
8	The Coming Woman: A Novel Based on	£17.93	Three	In stock
9	The Boys in the Boat: Nine Americans a	£22.60	Four	In stock
10	The Black Maria	£52.15	One	In stock
11	Starving Hearts (Triangular Trade Trilog	£3.59	Two	In stock
12	Shakespeare's Sonnets	£20.66	Four	In stock
13	Set Me Free	£17.46	Five	In stock
14	Scott Pilgrim's Precious Little Life (Scot	£52.29	Five	In stock
15	Rip It Up and Start Again	£35.02	Five	In stock
16	Our Band Could Be Your Life: Scenes fr	£57.25	Three	In stock
17	Olio	£23.88	One	In stock
18	Mesaeron: The Best Science Fiction St	£37.59	One	In stock
19	Libertarianism for Beginners	£51.3	Two	In stock
20	It's Only the Himalayas	£45.17	Two	In stock
21	In Her Wake	£12.84	One	In stock
22	How Music Works	£37.32	Two	In stock
23	Foodproof Preserving: A Guide to Smal	£23.2	Three	In stock
24	Chase Me (Paris Nights #2)	£25.27	Five	In stock
25	Black Dust	£34.53	Five	In stock
26	Birdsong: A Story in Pictures	£54.64	Three	In stock
27	America's Cradle of Quarterbacks: Wes	£22.30	Three	In stock
28	Aladdin and His Wonderful Lamp	£53.13	Three	In stock
29	Worlds Elsewhere: Journeys Around SH	£40.30	Five	In stock
30	Wall and Piece	£44.18	Four	In stock
31	The Four Agreements: A Practical Guide	£17.66	Five	In stock
32	The Five Love Languages: How to Expre	£31.05	Three	In stock
33	The Elephant Tree	£23.82	Five	In stock
34	The Bear and the Piano	£36.89	One	In stock
35	Sophie's World	£15.34	Five	In stock
36	Penny Mayble	£31.29	Three	In stock
37	Maude (1883-1993):She Grew Up with	£18.02	Two	In stock
38	In a Dark, Dark Wood	£19.63	One	In stock
39	Behind Closed Doors	£52.22	Four	In stock

Image 2: The beginning of the books_data.xlsx file in Excel, showing four columns: Title, Price (still stored as text with a currency symbol), Rating (still stored as words like 'Three', 'One'), and Availability (all 'In stock').



963	Old Records Never Die: One Man's Que	£55.66	Two	In stock
964	Off Sides (Off #1)	£39.45	Five	In stock
965	Of Mice and Men	£47.11	Two	In stock
966	Myriad (Prenor #1)	£58.75	Four	In stock
967	My Perfect Mistake (Over the Top #1)	£38.92	Two	In stock
968	Ms. Marvel, Vol. 1: No Normal (Ms. Mar	£39.39	Four	In stock
969	Meditations	£25.89	Two	In stock
970	Matilda	£28.34	One	In stock
971	Lost Among the Living	£27.70	Four	In stock
972	Lord of the Flies	£24.89	Three	In stock
973	Listen to Me (Fusion #1)	£58.99	Three	In stock
974	Kitchens of the Great Midwest	£57.20	Five	In stock
975	Jane Eyre	£38.43	Five	In stock
976	Imperfect Harmony	£34.74	Four	In stock
977	Icing (Aces Hockey #2)	£40.44	Four	In stock
978	Hawkeye, Vol. 1: My Life as a Weapon	£45.24	Three	In stock
979	Having the Barbarian's Baby (Ice Planet	£34.96	Four	In stock
980	Giant Days, Vol. 1 (Giant Days #1-4)	£56.76	Four	In stock
981	Fruits Basket, Vol. 1 (Fruits Basket #1)	£40.28	Five	In stock
982	Frankenstein	£38.00	Two	In stock
983	Forever Rockers (The Rocker #12)	£28.80	Three	In stock
984	Fighting Fate (Fighting #6)	£39.24	Three	In stock
985	Emma	£32.93	Two	In stock
986	Eat, Pray, Love	£51.32	Three	In stock
987	Deep Under (Walker Security #1)	£47.09	Five	In stock
988	Choosing Our Religion: The Spiritual Un	£28.42	Four	In stock
989	Charlie and the Chocolate Factory (Cha	£22.85	Three	In stock
990	Charity's Cross (Charles Towne Belles	£41.24	One	In stock
991	Bright Lines	£39.07	Five	In stock
992	Bridget Jones's Diary (Bridget Jones #1)	£29.92	One	In stock
993	Bounty (Colorado Mountain #7)	£37.26	Four	In stock
994	Blood Defense (Samantha Brinkman #1)	£20.30	Three	In stock
995	Bleach, Vol. 1: Strawberry and the Soul	£34.65	Five	In stock
996	Beyond Good and Evil	£43.38	One	In stock
997	Alice in Wonderland (Alice's Adventure	£55.53	One	In stock
998	Ajin: Demi-Human, Volume 1 (Ajin: Dei	£57.06	Four	In stock
999	A Spy's Devotion (The Regency Spies of	£16.97	Five	In stock
1000	1st to Die (Women's Murder Club #1)	£53.96	One	In stock
1001	1,000 Places to See Before You Die	£26.08	Five	In stock
1002				

Image 3: The end of the data (records 963 to 1000), confirming a total of 1,000 books extracted, and showing that the same issue pattern (Price as text, Rating as words) continues through to the last record.

Issues identified and how they were resolved:

- **Price stored as text** — The Price column contained a currency symbol embedded in the text (e.g. '£51.77'), so it could not be used directly in calculations (SUM, AVERAGE). **Fix:** a new numeric column 'Price (GBP)' was created by removing the '£' symbol and converting the value to a float.

- **Rating stored as words** — The Rating column contained words (One, Two, Three, Four, Five) instead of numbers. **Fix:** a new column 'Rating (1-5)' was created with a direct mapping (One=1 ... Five=5), while the original word column was kept for reference.
- **Availability has no variation** — All 1,000 books have a single value: 'In stock', because the practice site (toscrap.com) does not reflect real stock status. This column has limited analytical value but was kept for transparency.
- **Duplicate record** — The book 'The Star-Touched Queen' appears twice with different prices (£46.02 and £32.30) but the same Rating (Five). This may be due to the practice/demo nature of the source site. Both records were kept in the data and flagged in this report.
- **Currency symbol displaying incorrectly (Â£) in some tools** — In some cases, certain applications (or earlier screenshots) displayed '£' as 'Â£' — a common 'mojibake' issue that occurs when a program reads UTF-8 text using a different encoding (e.g. Latin-1). The actual .xlsx file contains the correct '£' symbol; this was purely a display issue.
- **Extra whitespace** — Some Title and Availability text values had leading/trailing spaces — these were trimmed.

The cleaning results are stored in a new file, `books_data_CLEANED.xlsx`, with three sheets: **Data (Safi)** — the cleaned data as an Excel Table; **Muhtasari** — overall statistics via formulas (COUNTA, AVERAGE, COUNTIF); and **Maelezo ya Usafishaji** — a full write-up of the issues found and how they were resolved.

7. Key Insights & Findings

- Total books analyzed: **1,000**
- Average price across all books: **£35.07** (lowest £10.00, highest £59.99)
- Average rating across all books: **2.92 out of 5**
- Rating distribution: 1★ = 226 books, 2★ = 196, 3★ = 203, 4★ = 179, 5★ = 196 — a fairly even spread (no single rating dominates)
- All books (100%) are listed as 'In stock' — this column cannot be used to differentiate availability
- Only 1 duplicate (2 records) was found by title among the 1,000 books

8. Recommendations

- If this were a real commercial project, the source data should include an *Availability* column with a real stock count rather than a static value, so it carries genuine business meaning.
- Adding a *Category/Genre* column (available on each book's individual page) would deepen the analysis — e.g. comparing prices across book categories.
- Building automatic duplicate detection into the scraping script itself, before saving the data, would reduce the manual cleaning workload.
- Explicitly using 'encoding=utf-8' when writing/reading files would prevent currency symbols from displaying incorrectly in some applications.

9. Conclusion

This project demonstrated a complete data cycle — from **extraction** via Python and web scraping, to **cleaning and analysis** in Excel. The main challenges encountered are typical of web-scraped data (data types stored as text instead of numbers, a column with no variation, and duplicate records), and all were handled transparently. Skills built: web scraping, data cleaning, and professional presentation of results.

10. Files & Links

- GitHub Repository: <https://github.com/bramwelsamwel16>
- Portfolio Website: <https://bramwelsamwel16.github.io/portfolio/>
- Files: scraper.py, books_data.xlsx (raw), books_data_CLEANED.xlsx, README.md